

NLP-based Psychological Diagnosis Diary Service

Team C (Running Machine)

1 Introduction

Mental health problems are becoming increasingly prominent in modern society. According to the Korea Suicide Prevention Foundation’s 2022 statistics, South Korea had a suicide rate of 26.6 per 100,000 people, the highest among OECD member countries [1]. These mental health issues not only diminish the quality of life for individuals but also impose a significant burden on society. Traditional mental health management methods, such as professional counseling or treatment, often require substantial time and financial resources. Additionally, many individuals hesitate to seek help due to the social stigma surrounding mental health issues [2].

To address these challenges, this project proposes a new method for automatically analyzing and managing mental health using Natural Language Processing (NLP) technology through a diary service. By analyzing users’ written diaries, NLP can detect emotional states such as depression, anger, and sadness. Based on these results, the system will provide suggestions for self-care or professional counseling. This service aims to detect and prevent mental health issues early, reduce the burden on professional counseling services, and protect users’ privacy. Ultimately, it is expected to contribute to improving mental health at a national level and enhancing the quality of life for individuals.

2 Motivation

Mental health issues in South Korea have reached a critical level. According to the Ministry of Health and Welfare in 2022, one in four South Koreans will experience a mental illness at least once in their lifetime [3]. The prevalence of depression is particularly concerning, with a rate of 6.7%, exceeding the OECD average of 5.5%. Furthermore, South Korea continues to hold the highest suicide rate among OECD countries, with 26.9 suicides per 100,000 people in 2022 [4]. These high levels of mental health problems not only degrade individual quality of life but also result in increased social costs and far-reaching impacts.

Traditional psychological counseling and treatment methods have their limitations. Professional counseling is often costly and involves long waiting periods, while social stigma associated with mental illness prevents many people from seeking help [2]. Consequently, there is a growing need for alternative solutions that can identify and prevent mental health problems early on.

The rapid advancement of Natural Language Processing (NLP) technology now makes it possible to assess emotional states through text data. NLP-based psychological analysis can detect emotions such as depression, anger, and anxiety by analyzing text written by users. This allows for early detection and intervention in mental health issues, providing opportunities for self-care without the need for professional counseling. Additionally, the anonymity provided by such a service reduces the burden of social stigma, making it easier for individuals to address their mental health concerns.

Thus, an NLP-based psychological state analysis service is expected to overcome the limitations of traditional methods and significantly contribute to improving national mental health. This project proposes a new approach that allows anyone to conveniently manage their mental health.

3 Problem Statement

Despite growing awareness of mental health issues, there remains a significant gap in the early detection and management of psychological conditions. Existing solutions, such as professional counseling, often require substantial time and financial resources, and are inaccessible to a large portion of the population due to social stigma and logistical challenges. Moreover, many individuals hesitate to seek help, leading to delayed diagnoses and worsened conditions.

To address these gaps, this project aims to leverage NLP technology to create a diary-based mental health monitoring service. This service provides an accessible and private way to analyze emotional states and detect early signs of mental health problems, offering personalized suggestions and resources for self-care or professional intervention. By making mental health monitoring more accessible and reducing stigma, this project hopes to enable timely interventions and improve overall mental well-being.

4 Related Work

Technological approaches to mental health monitoring have advanced significantly in recent years, particularly with the rise of Natural Language Processing (NLP) and Machine Learning (ML) applications. Below, we review several relevant works that have contributed to this field.

4.1 NLP in Mental Health

Natural language processing has increasingly been used to analyze text data for mental health diagnostics. For instance, the study titled “Linguistic Patterns for Depression Detection in Social Media” (2020) demonstrated how linguistic patterns in social media posts could help identify users at risk of depression. The system used NLP algorithms to detect key emotional indicators in text, providing an automated method for mental health monitoring. Similarly, the paper “Sentiment Analysis for Predicting Depressive Symptoms in Diaries” (2016)

applied sentiment analysis to user diaries to predict the onset of depressive symptoms.

4.2 Longitudinal Mental Health Monitoring

Recent research has emphasized the importance of longitudinal data in understanding mental health. For example, the paper titled “The Benefits of Longitudinal Mental Health Monitoring” (2019) explored the advantages of tracking mental health symptoms over time, highlighting that daily monitoring can improve the accuracy of mental health diagnoses. Our approach leverages this by using cumulative data from daily diary entries, enabling more precise detection of emotional and psychological patterns.

4.3 Emotion Detection and Sentiment Analysis

Emotion detection plays a crucial role in understanding an individual’s mental state. Recent NLP models such as BERT and GPT have shown considerable success in detecting sentiment and emotions in text data. For instance, the paper “BERT: State-of-the-Art Emotion Classification” (2019) presented BERT, which achieved state-of-the-art results in emotion classification tasks. Our system leverages such models to detect subtle changes in users’ emotional states based on their diary entries, contributing to more accurate psychological diagnoses.

5 Technology Background

5.1 BERT (Bidirectional Encoder Representations from Transformers)

BERT is an NLP model based on the transformer model, which uses self-attention mechanisms to process and understand the relationships between all words in a sentence, regardless of their positions. Transformers enable BERT to handle long-range dependencies in text more efficiently than traditional models. BERT further improves upon transformers by using a bidirectional encoder, which captures context by considering both preceding and following words in a sentence. This bidirectional processing allows BERT to understand words in a more contextually appropriate way compared to traditional unidirectional models.

5.2 KoBERT (Korean Bidirectional Encoder Representations from Transformers)

BERT faces challenges in multilingual settings, as it performs better when trained on a single language. KoBERT was developed to improve performance in the Korean language. It is pre-trained for Korean and uses an optimized

vocabulary and tokenizer specifically for the language. KoBERT is trained on a large corpus consisting of millions of Korean sentences from sources such as Wikipedia and news articles, reflecting the irregular linguistic variations of Korean. KoBERT achieved a more than 2.6% performance improvement with 27% fewer tokens compared to previous methods.

6 Planning in Detail

6.1 Role Distribution

This project is divided into two main parts: AI and web development. The AI part includes data collection, data preprocessing, and AI model implementation. Web development can be divided into three parts: UX/UI design, front-end, and back-end development. As the project progresses, roles may be further subdivided, and the assignments of individual team members may change.

Name	Role Description
Yehdarm Kim	UX/UI design
Nayeong Kim	AI implementation
Junhyun Park	Back-end development
Giho Bang	Front-end development

6.2 Development Plan

The project schedule spans a total of 15 weeks, with each subtask divided into two-week intervals. The timeline may be adjusted depending on the situation and project progress.

Weeks	2-3	4-5	6-7	8-9	10-11	12-13	14-15
Topic Ideation	O						
Data Collection		O	O				
AI Model Design		O	O	O			
Data Preprocessing			O	O			
Model & Algorithm Implementation			O	O			
UX/UI Design & Implementation				O	O	O	
Beta Service						O	O
Feedback & Revision						O	O

References

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